Pilot outline

PLANNED · NOT YET CARRIED OUT

A proposed 12-month pilot on 3 buses. It is planned, not yet carried out: the steps and measures below describe what the pilot would do and how success would be judged.

Monitor first → Human-reviewed alerts → HVAC control only with separate approval

What the current demo shows

- One simulated scenario of a city-bus cabin (CFD)
- Precomputed AI reconstructions, compared with the CFD reference
- Zone summaries and a click-to-inspect location readout
- An illustrative operator review workflow

Simulated: One steady CFD simulation of a city-bus cabin: 7 people, sun, 35 °C outside, 18 °C supply air. **Precomputed:** AI reconstructions (three neural networks) computed in advance from part of that simulation. Nothing is predicted live. **Illustrative:** The exterior body, wheels and mirrors are an illustration so the bus is recognisable; cabin fields come from the CFD dataset. The operator review list and the proposed sensor locations are illustrative too. **Planned:** Sensors on buses, monitoring, staff-reviewed alerts and their evaluation are planned for a pilot. They are not part of this demo.

Phases (12 months)

When	Phase	Activities	Milestone
Months 1–3	Build and test	Sensor nodes and a monitoring dashboard; check readings and fix problems.	Prototype ready
Months 4–6	Test on buses	Install on 3 buses; record trips and feedback.	Pilot dataset
Months 7–9	Test useful alerts	Create alerts and advice; test them with bus staff, who review every suggestion.	Alerts tested
Months 10–12	Evaluate and plan	Review results and costs; plan larger fleet tests.	Rollout decision

Proposed success measures

Measure	How it would be measured
Sensor uptime and coverage	Share of trip time with valid readings, per zone
Zone temperature differences	Measured differences between driver, front, middle and rear zones
Staff response time	Time from an alert to a staff review or action, from the review log
Passenger feedback	Short, accessible surveys on matched trips
HVAC energy per route-km	Only if energy metering is available on the pilot buses
Air-quality exceedance time	Only once air-quality sensors and a threshold protocol are agreed

These are proposed success measures. No results exist yet, so no values are shown.

What a pilot needs from an operator

- Three buses and permission to install removable equipment
- Installation windows outside service hours
- Access to trip schedules and, if available, HVAC and energy data
- Bus staff willing to review alerts and give feedback

Safeguards

- **Passenger privacy:** Aggregated occupancy only; no personal identification.
- **Vehicle safety:** No automatic HVAC control in the pilot; any later intervention keeps manual override.
- **Measurement quality:** Calibration, audit trails and documented thresholds before any claim about air quality or comfort.
- **Service disruption:** Removable equipment, installed off duty.

Illustrative cost estimate (not a price quote)

Item	Estimate
Equipment and calibration	USD 2,293
Cloud, visits and surveys	USD 2,158
Development labour (estimated)	USD 5,660
Contingency	USD 1,517
Total (estimate)	**USD 11,628**

- From the team's draft pilot plan for 3 buses over 12 months; includes estimated development labour.
- Potential cash need is about USD 5,968 if the development labour (USD 5,660) is provided in kind.
- An estimate for discussion only: it will change with the operator's requirements, and it is not an offer or a return-on-investment claim.